

Concordia University

Dept. of Computer Science & Software Engineering
COMP 353 – Databases — Summer 2026

Warm-up Project Report

A Simple Database Application for the Country Soccer Club System (CSCS)

Group Account: wqc353_1

Due: July 26, 2026

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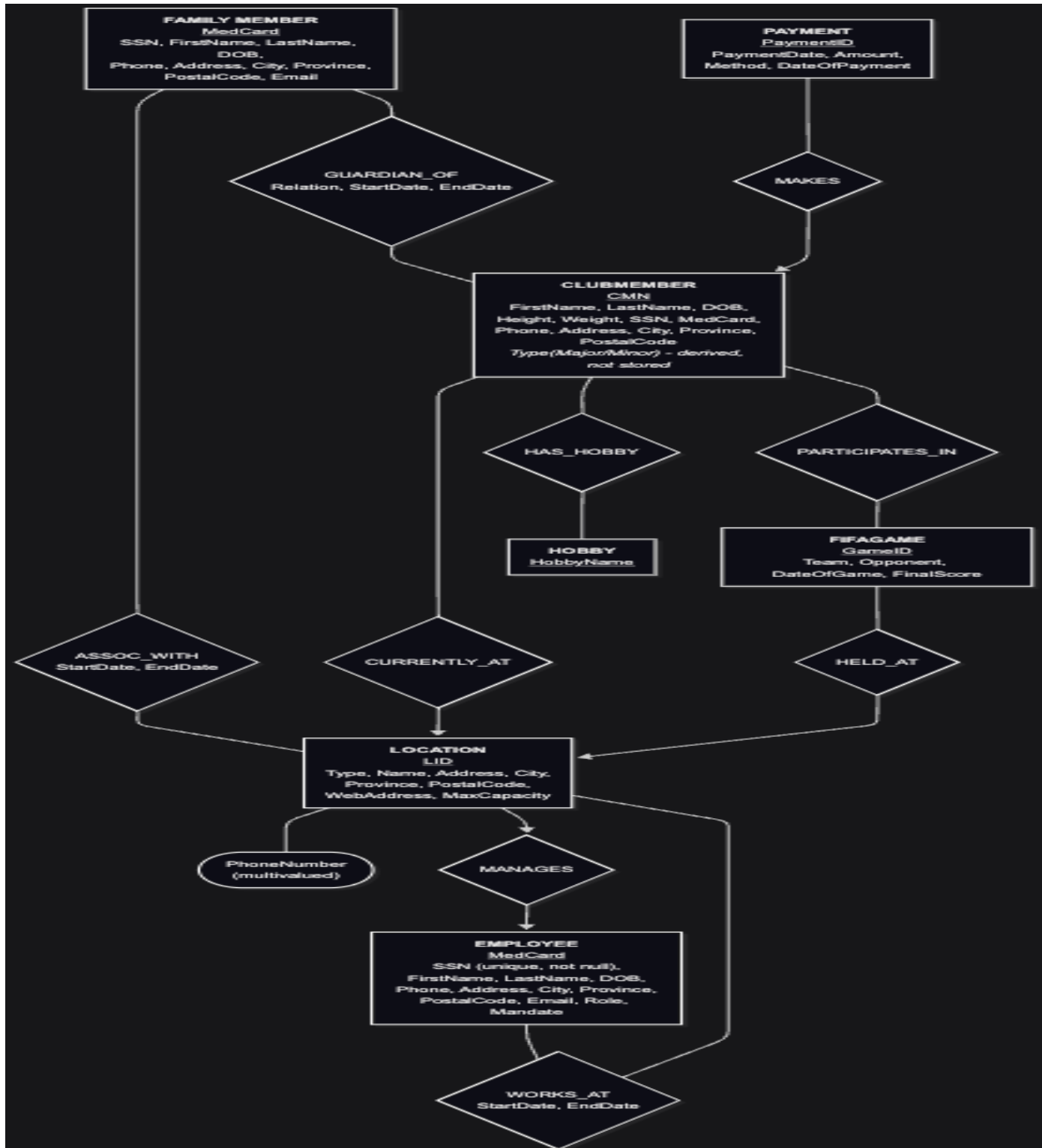
Repository: [cscs-db-353](#)

GUI DataBase Link: [wqc353.encs.concordia](#)

The forms are signed and attached at the end of the report

1. Design — E/R Diagram

The Country Soccer Club System (CSCS) is modeled above uses the *Entity-Relationship model*.



LEGEND

- Rectangles represent entity sets;
- Diamonds represent **relationship sets**;
- Ovals represent **multivalued attributes**;
- **Primary key attributes are marked (PK)**;
- Arrowheads point to the "1" side of a relationship; and
- Plain line marks the "many" side.

Key Design Decisions

- The Personnel-to-Location and FamilyMember-to-Location associations are time-dependent (WORKS_AT, ASSOC_WITH); thus, each has its own associative table that has its own StartDate/EndDate attributes, as opposed to a foreign key column on the entity. That is because a person may change locations and all of this needs to be tracked.
- The association between ClubMember and FamilyMember (GUARDIAN_OF) is time-dependent and is M:N – that is, a minor club member can have different family members as his guardians at different periods of time, and one and the same family member can be the guardian of multiple club members.
- The FIFAGAME entity is a strong entity which has M:N association with the ClubMember entity through the PARTICIPATES_IN relationship. In this way, we can avoid repetition of the game information (opponent, score, date) for each individual participant
- ClubMember.Type (Major/Minor) and "active/inactive" attribute (membership status) are derived attributes, computed on demand based on DateOfBirth and payment history attributes, respectively. We do not want to store those in the database; otherwise, the data will get out-of-date.
- The CMN (membership number) attribute is a single, system-wide auto-increment attribute serving as a primary key.

2. Database Schema — DDL, Keys, and Relationships

The above E/R diagram has been transformed into relational forms using the standard transformation rules that include: An entity set is transformed to a table; A many-to-many or temporal relationship is transformed into a junction table which has either a composite key or surrogate key; A one-to-many relationship is transformed into a foreign key at the “many” end; and

2.1 Primary Keys

Table	Primary Key
Location	location_id
LocationPhone	(location_id, phone_number)
Personnel	personnel_id
PersonnelAssignment	assignment_id
FamilyMember	family_member_id
FamilyMemberAssignment	assignment_id
ClubMember	membership_number (AUTO_INCREMENT)
ClubMemberFamilyRelation	relation_id
Hobby	hobby_id
ClubMemberHobby	(membership_number, hobby_id)
Payment	payment_id
FIFAGame	game_id
FIFAParticipation	(game_id, membership_number)

2.2 Foreign Keys

Table	Foreign Key	References
PersonnelAssignment	personnel_id	Personnel
PersonnelAssignment	location_id	Location
FamilyMemberAssignment	family_member_id	FamilyMember
FamilyMemberAssignment	location_id	Location
ClubMember	location_id	Location
ClubMemberFamilyRelation	membership_number	ClubMember
ClubMemberFamilyRelation	family_member_id	FamilyMember
ClubMemberHobby	membership_number	ClubMember
ClubMemberHobby	hobby_id	Hobby
Payment	membership_number	ClubMember
FIFAParticipation	membership_number	ClubMember
FIFAParticipation	game_id	FIFAGame
LocationPhone	location_id	Location

2.3 Constraints

- Personnel.ssn and FamilyMember.ssn are NOT NULL and UNIQUE, per the specification.
- Medicare numbers are UNIQUE for Personnel and FamilyMember (no two people share a Medicare card number).
- Personnel role is restricted to an ENUM of the roles named in the specification (General Manager, Deputy Manager, Treasurer, Secretary, Administrator, Captain, Coach, Assistant Coach, Other).
- Personnel mandate is restricted to ENUM('Volunteer', 'Salaried').
- Payment method is restricted to ENUM('Cash', 'Debit', 'Credit').
- ClubMember.membership_number is a global AUTO_INCREMENT primary key, unique across all locations.
- Foreign keys ensure referential integrity everywhere — no orphaned rows for assignments, payments, hobbies or game participation.
- DATE type is used for all date columns; DECIMAL(10,2) type is used for money columns; DECIMAL(5,2) type is used for height/weight columns.

2.4 Data Types Used

Data Type	Purpose
INT	Primary and foreign keys
VARCHAR	Names, addresses, emails, phone numbers, SSN, Medicare numbers
DATE	Birth dates, assignment dates, payment dates, game dates
DECIMAL(10,2)	Payment amounts
DECIMAL(5,2)	Height and weight
ENUM	Personnel role, mandate, payment method, location type, relationship type

2.5 Full Schema DDL (01_schema.sql)

```
-- Country Soccer Club System

USE wqc353_1;

CREATE TABLE Location (
    location_id INT AUTO_INCREMENT PRIMARY KEY,
    location_type ENUM('Head', 'Branch') NOT NULL,
    name VARCHAR(100) NOT NULL UNIQUE,
    address VARCHAR(150) NOT NULL,
    city VARCHAR(60) NOT NULL,
    province CHAR(2) NOT NULL,
    postal_code VARCHAR(10) NOT NULL,
    web_address VARCHAR(200),
    capacity INT NOT NULL
);

CREATE TABLE LocationPhone (
    location_id INT NOT NULL,
    phone_number VARCHAR(20) NOT NULL,
    PRIMARY KEY (location_id, phone_number),
    FOREIGN KEY (location_id) REFERENCES Location(location_id)
    ON DELETE CASCADE
);

CREATE TABLE Personnel (
    personnel_id INT AUTO_INCREMENT PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    date_of_birth DATE NOT NULL,
    ssn VARCHAR(15) NOT NULL UNIQUE,
    medicare_number VARCHAR(20) UNIQUE,
    phone_number VARCHAR(20),
    address VARCHAR(150),
    city VARCHAR(60),
    province CHAR(2),
    postal_code VARCHAR(10),
    email VARCHAR(100),
    role ENUM(
        'General Manager',
        'Deputy Manager',
        'Treasurer',
        'Secretary',
        'Administrator',
        'Captain',
        'Coach',
        'Assistant Coach',
        'Other'
    ) NOT NULL,
    mandate ENUM('Volunteer', 'Salaried') NOT NULL
);

CREATE TABLE PersonnelAssignment (
    assignment_id INT AUTO_INCREMENT PRIMARY KEY,
    personnel_id INT NOT NULL,
    location_id INT NOT NULL,
    start_date DATE NOT NULL,
    end_date DATE,
    UNIQUE (personnel_id, location_id, start_date),
    FOREIGN KEY (personnel_id) REFERENCES Personnel(personnel_id),
    FOREIGN KEY (location_id) REFERENCES Location(location_id),
    CHECK (end_date IS NULL OR end_date >= start_date)
);
```

```

CREATE TABLE FamilyMember (
    family_member_id INT AUTO_INCREMENT PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    date_of_birth DATE NOT NULL,
    ssn VARCHAR(15) NOT NULL UNIQUE,
    medicare_number VARCHAR(20) UNIQUE,
    phone_number VARCHAR(20),
    address VARCHAR(150),
    city VARCHAR(60),
    province CHAR(2),
    postal_code VARCHAR(10),
    email VARCHAR(100)
);

CREATE TABLE FamilyMemberAssignment (
    assignment_id INT AUTO_INCREMENT PRIMARY KEY,
    family_member_id INT NOT NULL,
    location_id INT NOT NULL,
    start_date DATE NOT NULL,
    end_date DATE,
    UNIQUE (family_member_id, location_id, start_date),
    FOREIGN KEY (family_member_id) REFERENCES FamilyMember(family_member_id),
    FOREIGN KEY (location_id) REFERENCES Location(location_id),
    CHECK (end_date IS NULL OR end_date >= start_date)
);

CREATE TABLE ClubMember (
    membership_number INT AUTO_INCREMENT PRIMARY KEY,
    location_id INT NOT NULL,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    date_of_birth DATE NOT NULL,
    registration_date DATE NOT NULL,
    height_cm DECIMAL(5,2),
    weight_kg DECIMAL(5,2),
    ssn VARCHAR(15) UNIQUE,
    medicare_number VARCHAR(20) UNIQUE,
    phone_number VARCHAR(20),
    address VARCHAR(150),
    city VARCHAR(60),
    province CHAR(2),
    postal_code VARCHAR(10),
    FOREIGN KEY (location_id) REFERENCES Location(location_id)
);

CREATE TABLE ClubMemberFamilyRelation (
    relation_id INT AUTO_INCREMENT PRIMARY KEY,
    membership_number INT NOT NULL,
    family_member_id INT NOT NULL,
    relationship_type ENUM(
        'Father', 'Mother', 'Grandfather', 'Grandmother',
        'Tutor', 'Partner', 'Friend', 'Other'
    ) NOT NULL,
    start_date DATE NOT NULL,
    end_date DATE,
    UNIQUE (membership_number, family_member_id, start_date),
    FOREIGN KEY (membership_number)
        REFERENCES ClubMember(membership_number),
    FOREIGN KEY (family_member_id)
        REFERENCES FamilyMember(family_member_id),
    CHECK (end_date IS NULL OR end_date >= start_date)
);

CREATE TABLE Hobby (
    hobby_id INT AUTO_INCREMENT PRIMARY KEY,
    hobby_name VARCHAR(50) NOT NULL UNIQUE
);

CREATE TABLE ClubMemberHobby (
    membership_number INT NOT NULL,
    hobby_id INT NOT NULL,
    PRIMARY KEY (membership_number, hobby_id),
    FOREIGN KEY (membership_number)
        REFERENCES ClubMember(membership_number)
        ON DELETE CASCADE,

```

```

FOREIGN KEY (hobby_id)
  REFERENCES Hobby(hobby_id)
);

CREATE TABLE Payment (
  payment_id INT AUTO_INCREMENT PRIMARY KEY,
  membership_number INT NOT NULL,
  payment_date DATE NOT NULL,
  amount DECIMAL(10,2) NOT NULL,
  payment_method ENUM('Cash', 'Debit', 'Credit') NOT NULL,
  membership_year YEAR NOT NULL,
  installment_number INT NOT NULL DEFAULT 1,
  UNIQUE (membership_number, membership_year, installment_number),
  FOREIGN KEY (membership_number)
    REFERENCES ClubMember(membership_number),
  CHECK (amount > 0),
  CHECK (installment_number BETWEEN 1 AND 4)
);

CREATE TABLE FIFAGame (
  game_id INT AUTO_INCREMENT PRIMARY KEY,
  location_id INT NOT NULL,
  team_name VARCHAR(100) NOT NULL,
  opponent_name VARCHAR(100) NOT NULL,
  game_date DATE NOT NULL,
  team_score INT NOT NULL,
  opponent_score INT NOT NULL,
  FOREIGN KEY (location_id) REFERENCES Location(location_id)
);

CREATE TABLE FIFAParticipation (
  game_id INT NOT NULL,
  membership_number INT NOT NULL,
  PRIMARY KEY (game_id, membership_number),
  FOREIGN KEY (game_id)
    REFERENCES FIFAGame(game_id)
    ON DELETE CASCADE,
  FOREIGN KEY (membership_number)
    REFERENCES ClubMember(membership_number)
);

```

3. SQL Queries and Transactions

The eight required queries below are implemented in `sql/03_queries.sql`. Each corresponds to one item (i–viii) from the project description.

```
-- Query (i): Complete details for every location
WITH Phones AS (
  SELECT location_id, GROUP_CONCAT(phone_number SEPARATOR ', ') AS phone_numbers
  FROM LocationPhone
  GROUP BY location_id
),
GM AS (
  SELECT pa.location_id,
         CONCAT(p.first_name, ' ', p.last_name) AS gm_name
  FROM PersonnelAssignment pa
  JOIN Personnel p ON p.personnel_id = pa.personnel_id
  WHERE p.role = 'General Manager'
  AND pa.end_date IS NULL
),
PersonnelCount AS (
  SELECT location_id, COUNT(DISTINCT personnel_id) AS num_personnel
  FROM PersonnelAssignment
  WHERE end_date IS NULL
  GROUP BY location_id
),
MemberCount AS (
  SELECT location_id, COUNT(*) AS num_club_members
  FROM ClubMember
  GROUP BY location_id
),
FifaMemberCount AS (
  SELECT cm.location_id, COUNT(DISTINCT fp.membership_number) AS num_fifa_players
  FROM ClubMember cm
  JOIN FIFAParticipation fp ON fp.membership_number = cm.membership_number
  GROUP BY cm.location_id
)
```

location_name	address	city	province	postal_code	phone_numbers	web_address	location_type	capacity	general_manager	num_personnel	num_club_members	num_fifa_players
CSCS North Branch	200 Boulevard Laurentien	Saint-Laurent	QC	H4M2M2	450-555-0400	https://north.cscs.ca	Branch	175	NULL	1	0	0
CSCS South Branch	300 Rue Taschereau	Brossard	QC	J4Y1A1	450-555-0500	https://south.cscs.ca	Branch	180	NULL	1	0	0
CSCS Central Branch	500 Rue Sainte-Anne	Varenes	QC	J3X1P7	450-555-0600	https://central.cscs.ca	Branch	160	NULL	1	0	0
CSCS Plateau Branch	80 Avenue du Mont-Royal E	Montreal	QC	H2T1P1	514-555-0700	https://plateau.cscs.ca	Branch	120	NULL	0	0	0
CSCS Verdun Branch	5 Rue de Verdun	Montreal	QC	H4G2M5	514-555-0800	https://verdun.cscs.ca	Branch	130	NULL	0	0	0
CSCS Rive-Nord Branch	750 Chemin du Bas-de-la-Riviere	Terrebonne	QC	J6W1A3	450-555-0900	https://rive-nord.cscs.ca	Branch	140	NULL	0	0	0
CSCS Anjou Branch	7000 Rue Sherbrooke E	Montreal	QC	H1K1C3	514-566-0120	https://anjou.cscs.ca	Branch	110	NULL	0	0	0
CSCS West Branch	450 Boulevard Cur��-Labelle	Laval	QC	H7P2P2	450-555-0200	https://west.cscs.ca	Branch	200	NULL	1	3	3
CSCS East Branch	25 Rue Saint-Charles	Longueuil	QC	J4H1E5	450-555-0300	https://east.cscs.ca	Branch	150	NULL	2	3	2
CSCS Head Office	100 Rue Sainte-Catherine	Montreal	QC	H2K1K4	514-555-0100, 514-555-0101	https://cscs.ca	Head	500	Marc Tremblay	3	4	3

```
-- Query (ii): Major club members who played >= 1 FIFA game
```

```
WITH AgeCalc AS (
  SELECT membership_number,
         TIMESTAMPDIFF(YEAR, date_of_birth, CURDATE()) AS age
  FROM ClubMember
),
PrevYearPay AS (
  SELECT membership_number, SUM(amount) AS total_paid
  FROM Payment
  WHERE membership_year = YEAR(CURDATE()) - 1
  GROUP BY membership_number
),
GameCounts AS (
  SELECT membership_number, COUNT(*) AS num_games
  FROM FIFAParticipation
  GROUP BY membership_number
)
SELECT
  l.name AS location_name,
  cm.membership_number,
  cm.first_name,
  cm.last_name,
  a.age,
  cm.city,
  cm.province,
  CASE WHEN COALESCE(p.total_paid, 0) >= 200 THEN 'Active' ELSE 'Inactive' END AS status,
  g.num_games
FROM ClubMember cm
JOIN AgeCalc a ON a.membership_number = cm.membership_number
JOIN Location l ON l.location_id = cm.location_id
JOIN GameCounts g ON g.membership_number = cm.membership_number
LEFT JOIN PrevYearPay p ON p.membership_number = cm.membership_number
```

location_name	membership_number	first_name	last_name	age	city	province	status	num_games
CSCS West Branch	4	Camille	Dubois	21	Laval	QC	Active	3
CSCS East Branch	6	Laurie	Pelletier	18	Longueuil	QC	Inactive	3
CSCS Head Office	7	Gabriel	Morin	20	Montreal	QC	Inactive	3
CSCS Head Office	2	Sophie	B��langer	18	Montreal	QC	Inactive	4


```

-- Query (iii): Club members with >= 4 different hobbies

WITH AgeCalc AS (
    SELECT membership_number,
           TIMESTAMPDIFF(YEAR, date_of_birth, CURDATE()) AS age
    FROM ClubMember
),
PrevYearPay AS (
    SELECT membership_number, SUM(amount) AS total_paid
    FROM Payment
    WHERE membership_year = YEAR(CURDATE()) - 1
    GROUP BY membership_number
),
HobbyCounts AS (
    SELECT membership_number, COUNT(DISTINCT hobby_id) AS num_hobbies
    FROM ClubMemberHobby
    GROUP BY membership_number
    HAVING COUNT(DISTINCT hobby_id) >= 4
)
SELECT
    l.name AS location_name,
    cm.membership_number,
    cm.first_name,
    cm.last_name,
    a.age,
    cm.city,
    cm.province,
    CASE
        WHEN COALESCE(p.total_paid, 0) >= (CASE WHEN a.age >= 18 THEN 200 ELSE 100 END)
        THEN 'Active' ELSE 'Inactive'
    END AS status,
    h.num_hobbies
FROM ClubMember cm
JOIN AgeCalc a      ON a.membership_number = cm.membership_number
JOIN Location l     ON l.membership_number = cm.membership_number

```

location_name	membership_number	first_name	last_name	age	city	province	status	num_hobbies
CSCS West Branch	4	Camille	Dubois	21	Laval	QC	Active	5
CSCS East Branch	6	Laurie	Pelletier	18	Longueuil	QC	Inactive	5

```
-- QUERY-4 (iv) Get all major club members (18+) who have never participated in any FIFA game --
SELECT
    l.name AS location_name,
    cm.membership_number,
    cm.first_name,
    cm.last_name,
    TIMESTAMPDIFF(YEAR, cm.date_of_birth, CURDATE()) AS age,
    cm.city,
    cm.province,

    CASE
        WHEN COALESCE((
            SELECT SUM(p.amount)
            FROM Payment p
            WHERE p.membership_number = cm.membership_number
              AND p.membership_year = YEAR(CURDATE()) - 1
        ), 0) >=
        CASE
            WHEN TIMESTAMPDIFF(
                YEAR,
                cm.date_of_birth,
                STR_TO_DATE(
                    CONCAT(YEAR(CURDATE()) - 1, '-12-31'),
                    '%Y-%m-%d'
                )
            ) >= 18
            THEN 200
            ELSE 100
        END
        THEN 'Active'
        ELSE 'Inactive'
    END AS status
FROM ClubMember cm
JOIN Location l
    ON l.location_id = cm.location_id
...
```

location_name	membership_number	first_name	last_name	age	city	province	status
CSCS East Branch	10	Isabelle	Caron	24	Longueuil	QC	Inactive
CSCS Head Office	9	Alexandre	Gagnon	27	Montreal	QC	Active

```
-- QUERY-5 (v): total number of club members for every age
SELECT
    TIMESTAMPDIFF(YEAR, date_of_birth, CURDATE()) AS age,
    COUNT(*) AS total_club_members
FROM ClubMember
GROUP BY age
ORDER BY age DESC
```

age	total_club_members
27	1
24	1
21	1
20	1
18	2
16	1
14	1
13	2

-- QUERY-6 (vi): Major members who are also family members

```
SELECT
  cm.membership_number,
  CONCAT(cm.first_name, ' ', cm.last_name) AS member_name,
  child.membership_number AS associated_member_number,
  CONCAT(child.first_name, ' ', child.last_name) AS associated_member_name,
  child.date_of_birth AS associated_dob,
  child.ssn AS associated_ssn,
  child.medicare_number AS associated_medicare_number,
  child.phone_number AS associated_phone,
  child.address AS associated_address,
  child.city AS associated_city,
  child.province AS associated_province,
  child.postal_code AS associated_postal_code,
  r.relationship_type,
  CASE WHEN
    member_number = associated_member_number
  THEN
    member_name = associated_member_name
  ELSE
    member_name <> associated_member_name
  END AS is_family_member
FROM ClubMember cm
JOIN FamilyMember r ON cm.membership_number = cm.associated_member_number
WHERE r.relationship_type IN ('parent', 'spouse', 'partner')
```

member_number	member_name	associated_member_number	associated_member_name	associated_dob	associated_ssn	associated_medicare_number	associated_phone	associated_address	associated_city	associated_province	associated_postal_code
9	Alexandre Gagnon	1	Thomas Lavoie	2010-02-14	M100000001	LAV010021499	514-555-3001	15 Rue Sherbrooke	Montreal	QC	H2F6F6
10	Isabelle Caron	3	Maxime Girard	2011-11-02	M100000003	GIRA11110299	450-555-3003	27 Rue L'Ã@vis	Laval	QC	H7G7G7

-- QUERY-7 (vii): Fees, major members, 2023-2025

```
SELECT
  cm.membership_number,
  CONCAT(cm.first_name, ' ', cm.last_name) AS member_name,
  p.membership_year,
  SUM(p.amount) AS total_paid
FROM ClubMember cm
JOIN Payment p ON p.membership_number = cm.membership_number
WHERE TIMESTAMPDIFF(YEAR, cm.date_of_birth, CURDATE()) >= 18
  AND p.membership_year BETWEEN 2023 AND 2025
GROUP BY cm.membership_number, member_name, p.membership_year
ORDER BY cm.membership_number, p.membership_year
```

membership_number	member_name	membership_year	total_paid
2	Sophie BÃ©langer	2024	300.00
4	Camille Dubois	2024	300.00
4	Camille Dubois	2025	200.00
6	Laurie Pelletier	2024	300.00
6	Laurie Pelletier	2025	100.00
7	Gabriel Morin	2024	300.00
9	Alexandre Gagnon	2025	200.00
10	Isabelle Caron	2025	100.00

-- QUERY-8 (viii): Members with >=4 FIFA games (all members)

```
SELECT
  cm.membership_number,
  CONCAT(cm.first_name, ' ', cm.last_name) AS member_name,
  COUNT(fp.game_id) AS games_played
FROM ClubMember cm
JOIN FIFAParticipation fp ON fp.membership_number = cm.membership_number
GROUP BY cm.membership_number, member_name
HAVING COUNT(fp.game_id) >= 4
```

membership_number	member_name	games_played
1	Thomas Lavoie	4
2	Sophie BÃ©langer	4

4. Database Population

The database is populated using `sql/02_seed.sql`, inserted in an order that respects foreign-key dependencies (Location before ClubMember, ClubMember before Payment, etc.). The specification requires at least 10 representative tuples per table such that every query returns at least two rows.

4.1 Current Row Counts

Table	Row Count	Meets ≥ 10 requirement?
Location	10	Yes
LocationPhone	11	Yes
Personnel	10	Yes
PersonnelAssignment	10	Yes
FamilyMember	10	Yes
FamilyMemberAssignment	10	Yes
ClubMember	10	Yes
ClubMemberFamilyRelation	10	Yes
Hobby	10	Yes
ClubMemberHobby	18	Yes
Payment	18	Yes
FIFAGame	10	Yes
FIFAParticipation	29	Yes

4.2 Seed Data (02_seed.sql)

```
-- =====
-- CSCS (Country Soccer Club System) - Seed Data
-- Insert order respects FK dependencies:
-- Location -> LocationPhone
-- Personnel -> PersonnelAssignment
-- FamilyMember -> FamilyMemberAssignment
-- Location -> ClubMember -> ClubMemberFamilyRelation
-- Hobby -> ClubMemberHobby
-- ClubMember -> Payment
-- Location -> FIFAGame -> FIFAParticipation
-- =====

USE wqc353_1;

-- =====
-- Table: Location
-- =====
INSERT INTO Location (location_id, location_type, name, address, city, province, postal_code, web_address, capacity) VALUES
(1, 'Head', 'CSCS Head Office', '100 Rue Sainte-Catherine', 'Montreal', 'QC', 'H2X1K4', 'https://cscs.ca', 500),
(2, 'Branch', 'CSCS West Branch', '450 Boulevard Curé-Labelle', 'Laval', 'QC', 'H7P2P2', 'https://west.cscs.ca', 200),
(3, 'Branch', 'CSCS East Branch', '25 Rue Saint-Charles', 'Longueuil', 'QC', 'J4H1E5', 'https://east.cscs.ca', 150),
(4, 'Branch', 'CSCS North Branch', '200 Boulevard Laurentien', 'Saint-Laurent', 'QC', 'H4M2M2', 'https://north.cscs.ca', 175),
(5, 'Branch', 'CSCS South Branch', '300 Rue Taschereau', 'Brossard', 'QC', 'J4Y1A1', 'https://south.cscs.ca', 180),
(6, 'Branch', 'CSCS Central Branch', '500 Rue Sainte-Anne', 'Varennes', 'QC', 'J3X1P7', 'https://central.cscs.ca', 160),
(7, 'Branch', 'CSCS Plateau Branch', '80 Avenue du Mont-Royal E', 'Montreal', 'QC', 'H2T1P1', 'https://plateau.cscs.ca', 120),
(8, 'Branch', 'CSCS Verdun Branch', '5 Rue de Verdun', 'Montreal', 'QC', 'H4G2M5', 'https://verdun.cscs.ca', 130),
(9, 'Branch', 'CSCS Rive-Nord Branch', '750 Chemin du Bas-de-la-Riviere', 'Terrebonne', 'QC', 'J6W1A3', 'https://rivenord.cscs.ca', 140),
(10, 'Branch', 'CSCS Anjou Branch', '7000 Rue Sherbrooke E', 'Montreal', 'QC', 'H1K1C5', 'https://anjou.cscs.ca', 110);

-- =====
-- Table: LocationPhone
-- =====
INSERT INTO LocationPhone (location_id, phone_number) VALUES
(1, '514-555-0100'),
(1, '514-555-0101'),
(2, '450-555-0200'),
(3, '450-555-0300'),
(4, '450-555-0400'),
(5, '450-555-0500'),
(6, '450-555-0600'),
(7, '514-555-0700'),
(8, '514-555-0800'),
(9, '450-555-0900'),
(10, '514-566-0120');

-- =====
-- Table: Personnel
-- =====
INSERT INTO Personnel (personnel_id, first_name, last_name, date_of_birth, ssn, medicare_number, phone_number, address, city, province, postal_code, email, role, mandate) VALUES
(1, 'Marc', 'Tremblay', '1975-03-12', '111-111-111', 'TREM75031299', '514-555-1001', '12 Rue Principale', 'Montreal', 'QC', 'H1A1A1', 'marc.tremblay@cscs.ca', 'General Manager', 'Salaried'),
(2, 'Julie', 'Bouchard', '1982-07-04', '222-222-222', 'BOUC82070499', '514-555-1002', '34 Avenue du Parc', 'Montreal', 'QC', 'H2B2B2', 'julie.bouchard@cscs.ca', 'Treasurer', 'Salaried'),
(3, 'Simon', 'Roy', '1990-11-23', '333-333-333', 'ROYS90112399', '450-555-1003', '56 Rue des Érables', 'Laval', 'QC', 'H7C3C3', 'simon.roy@cscs.ca', 'Coach', 'Volunteer'),
(4, 'Nadia', 'Cormier', '1988-01-30', '444-444-444', 'CORM88013099', '450-555-1004', '78 Rue de la Gare', 'Longueuil', 'QC', 'J4D4D4', 'nadia.cormier@cscs.ca', 'Secretary', 'Volunteer'),
(5, 'Eric', 'Fortin', '1995-05-17', '555-555-555', 'FORT95051799', '514-555-1005', '90 Rue Saint-Denis', 'Montreal', 'QC', 'H2E5E5', 'eric.fortin@cscs.ca', 'Assistant Coach', 'Volunteer'),
(6, 'Luc', 'Gauthier', '1985-06-14', '601-111-111', 'GAUT85061499', '514-555-1006', '22 Rue Masson', 'Montreal', 'QC', 'H2S2S2', 'luc.gauthier@cscs.ca', 'Deputy Manager', 'Salaried'),
(7, 'Isabeau', 'Pelletier', '1992-08-20', '602-111-111', 'PELL92082099', '514-555-1007', '63 Rue Beaubien', 'Montreal', 'QC', 'H2G3G3', 'isabeau.pelletier@cscs.ca', 'Captain', 'Volunteer'),
(8, 'Francois', 'Lapointe', '1987-04-05', '603-111-111', 'LAP087040599', '450-555-1008', '15 Rue Joliette', 'Brossard', 'QC', 'J4X4X4', 'francois.lapointe@cscs.ca', 'Administrator', 'Salaried'),
(9, 'Marie-Eve', 'Ouellet', '1993-12-11', '604-111-111', 'OUEL93121199', '450-555-1009', '88 Rue des Saules', 'Terrebonne', 'QC', 'J6W2W2', 'marieeve.ouellet@cscs.ca', 'Coach', 'Salaried'),
(10, 'David', 'Cloutier', '1980-09-03', '605-111-111', 'CLOU80090399', '514-555-1010', '44 Rue Notre-Dame O', 'Montreal', 'QC', 'H3C3C3', 'david.cloutier@cscs.ca', 'Secretary', 'Volunteer');

-- =====
-- Table: PersonnelAssignment
-- =====
INSERT INTO PersonnelAssignment (assignment_id, personnel_id, location_id, start_date, end_date) VALUES
(1, 1, 1, '2019-01-01', NULL), -- current GM at Head office
(2, 2, 1, '2020-06-01', NULL), -- current Treasurer at Head office
(3, 3, 2, '2021-03-15', NULL), -- current Coach at West branch
(4, 4, 3, '2022-02-01', NULL), -- current Secretary at East branch
(5, 5, 2, '2018-09-01', '2021-03-01'), -- past Assistant Coach assignment at West branch
(6, 5, 3, '2021-03-16', NULL), -- same person, now current at East branch
(7, 6, 1, '2022-01-10', NULL), -- Luc: Deputy Manager at Head office
(8, 7, 4, '2023-03-01', NULL), -- Isabeau: Captain at North branch
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(9, 8, 5, '2023-07-15', NULL), -- Francois: Administrator at South branch
(10, 9, 6, '2024-01-01', NULL); -- Marie-Eve: Coach at Central branch

-- =====
-- Table: FamilyMember
-- =====
INSERT INTO FamilyMember (family_member_id, first_name, last_name, date_of_birth, ssn, medicare_number, phone_number, address, city,
province, postal_code, email) VALUES
(1, 'Denise', 'Lavoie', '1970-04-08', '666-666-666', 'LAVO70040899', '514-555-2001', '15 Rue Sherbrooke', 'Montreal', 'QC', 'H2F6F6',
'denise.lavoie@email.com'),
(2, 'Patrick', 'Girard', '1972-09-19', '777-777-777', 'GIRA72091999', '450-555-2002', '27 Rue Lévis', 'Laval', 'QC', 'H7G7G7',
'patrick.girard@email.com'),
(3, 'Chantal', 'Bergeron', '1980-12-01', '888-888-888', 'BERG80120199', '450-555-2003', '39 Rue Fournier', 'Longueuil', 'QC', 'J4H8H8',
'chantal.bergeron@email.com'),
(4, 'Alain', 'Morin', '1978-06-25', '999-999-999', 'MORI78062599', '514-555-2004', '41 Rue Rachel', 'Montreal', 'QC', 'H2I9I9',
'alain.morin@email.com'),
(5, 'Robert', 'Tremblay', '1965-07-14', '101-010-101', 'TREM65071499', '514-555-2005', '72 Rue Beaubien', 'Montreal', 'QC', 'H2G3G3',
'robert.tremblay@email.com'),
(6, 'Martine', 'Gagnon', '1973-02-28', '202-020-202', 'GAGN73022899', '450-555-2006', '100 Rue du Boise', 'Laval', 'QC', 'H7R5R5',
'martine.gagnon@email.com'),
(7, 'Louis', 'Bedard', '1968-11-09', '303-030-303', 'BEDA68110999', '450-555-2007', '55 Rue des Tilleuls', 'Longueuil', 'QC', 'J4L7L7',
'louis.bedard@email.com'),
(8, 'Sylvie', 'Charron', '1975-05-03', '404-040-404', 'CHAR75050399', '514-555-2008', '18 Cote-des-Neiges', 'Montreal', 'QC', 'H3H1A1',
'sylvie.charron@email.com'),
(9, 'Pierre', 'Beaumont', '1969-08-16', '505-050-505', 'BEAU69081699', '450-555-2009', '33 Rue Edouard', 'Brossard', 'QC', 'J4Y2Y2',
'pierre.beaumont@email.com'),
(10, 'Helene', 'Leblanc', '1977-03-22', '606-060-606', 'LEBL77032299', '514-555-2010', '90 Boulevard Decarie', 'Montreal', 'QC', 'H4L3L3',
'helene.leblanc@email.com');

-- =====
-- Table: FamilyMemberAssignment
-- =====
INSERT INTO FamilyMemberAssignment (assignment_id, family_member_id, location_id, start_date, end_date) VALUES
(1, 1, 1, '2020-01-10', NULL),
(2, 2, 2, '2021-04-01', NULL),
(3, 3, 3, '2022-05-15', NULL),
(4, 4, 1, '2019-08-01', '2023-01-01'),
(5, 5, 4, '2023-03-01', NULL),
(6, 6, 2, '2021-09-15', NULL),
(7, 7, 3, '2022-11-01', NULL),
(8, 8, 5, '2023-08-20', NULL),
(9, 9, 1, '2024-01-05', NULL),
(10, 10, 6, '2024-03-10', NULL);

-- =====
-- Table: ClubMember
-- =====
INSERT INTO ClubMember (membership_number, location_id, first_name, last_name, date_of_birth, registration_date, height_cm, weight_kg, ssn,
medicare_number, phone_number, address, city, province, postal_code) VALUES
(1, 1, 'Thomas', 'Lavoie', '2010-02-14', '2022-08-01', 155.00, 45.50, 'M100000001', 'LAVO10021499', '514-555-3001', '15 Rue Sherbrooke',
'Montreal', 'QC', 'H2F6F6'),
(2, 1, 'Sophie', 'Bélanger', '2008-06-30', '2021-09-05', 162.00, 52.00, 'M100000002', 'BELA08063099', '514-555-3002', '20 Rue Ontario',
'Montreal', 'QC', 'H2K1K1'),
(3, 2, 'Maxime', 'Girard', '2011-11-02', '2023-01-15', 148.00, 40.00, 'M100000003', 'GIRA11110299', '450-555-3003', '27 Rue Lévis', 'Laval',
'QC', 'H7G7G7'),
(4, 2, 'Camille', 'Dubois', '2005-03-19', '2020-02-20', 170.00, 60.00, 'M100000004', 'DUBO05031999', '450-555-3004', '10 Rue des Pins',
'Laval', 'QC', 'H7L2L2'),
(5, 3, 'Antoine', 'Bergeron', '2012-09-08', '2023-06-10', 140.00, 36.00, 'M100000005', 'BERG12090899', '450-555-3005', '39 Rue Fournier',
'Longueuil', 'QC', 'J4H8H8'),
(6, 3, 'Laurie', 'Pelletier', '2007-12-25', '2021-11-01', 158.00, 48.00, 'M100000006', 'PELL07122599', '450-555-3006', '5 Rue Curé-
Poirier', 'Longueuil', 'QC', 'J4J3J3'),
(7, 1, 'Gabriel', 'Morin', '2006-04-17', '2020-09-01', 168.00, 58.00, 'M100000007', 'MORI06041799', '514-555-3007', '41 Rue Rachel',
'Montreal', 'QC', 'H2I9I9'),
(8, 2, 'Emma', 'Roy', '2013-01-05', '2024-01-20', 135.00, 32.00, 'M100000008', 'ROYE13010599', '450-555-3008', '56 Rue des Érables', 'Laval',
'QC', 'H7C3C3'),
(9, 1, 'Alexandre', 'Gagnon', '1998-08-14', '2023-05-10', 181.00, 76.00, 'M100000009', 'GAGN98081499', '514-555-3009', '72 Rue Saint-
Laurent', 'Montreal', 'QC', 'H2Y2Y2'),
(10, 3, 'Isabelle', 'Caron', '2001-10-22', '2024-02-15', 168.00, 59.00, 'M100000010', 'CARO01102299', '450-555-3010', '88 Rue Victoria',
'Longueuil', 'QC', 'J4K4K4');

-- =====
-- Table: ClubMemberFamilyRelation
-- =====
INSERT INTO ClubMemberFamilyRelation (relation_id, membership_number, family_member_id, relationship_type, start_date, end_date) VALUES
(1, 1, 1, 'Mother', '2022-08-01', NULL),
(2, 3, 2, 'Father', '2023-01-15', NULL),
(3, 5, 3, 'Mother', '2023-06-10', NULL),
(4, 7, 4, 'Father', '2020-09-01', NULL),
(5, 2, 5, 'Father', '2021-09-05', NULL),
(6, 4, 6, 'Mother', '2020-02-20', NULL),
(7, 6, 7, 'Father', '2021-11-01', NULL),
(8, 8, 8, 'Mother', '2024-01-20', NULL),
(9, 9, 9, 'Father', '2023-05-10', NULL),

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(10, 10, 10, 'Mother', '2024-02-15', NULL);

-- =====
-- Table: Hobby
-- =====
INSERT INTO Hobby (hobby_id, hobby_name) VALUES
(1, 'Reading'),
(2, 'Swimming'),
(3, 'Video Games'),
(4, 'Music'),
(5, 'Chess'),
(6, 'Painting'),
(7, 'Cycling'),
(8, 'Cooking'),
(9, 'Photography'),
(10, 'Dancing');

-- =====
-- Table: ClubMemberHobby
-- =====
INSERT INTO ClubMemberHobby (membership_number, hobby_id) VALUES
(1, 2), (1, 5), -- Thomas: Swimming, Chess
(2, 1), (2, 4), -- Sophie: Reading, Music
(3, 3), -- Maxime: Video Games
(4, 1), (4, 2), (4, 4), -- Camille: Reading, Swimming, Music
(5, 5), -- Antoine: Chess
(6, 3), (6, 1), -- Laurie: Video Games, Reading
(7, 2), -- Gabriel: Swimming
(8, 4); -- Emma: Music

-- =====
-- Table: Payment
-- =====
INSERT INTO Payment (payment_id, membership_number, payment_date, amount, payment_method, membership_year, installment_number) VALUES
-- Thomas: paid in full
(1, 1, '2024-08-01', 300.00, 'Debit', 2024, 1),
-- Sophie: paid in 2 installments
(2, 2, '2024-09-01', 150.00, 'Credit', 2024, 1),
(3, 2, '2024-11-01', 150.00, 'Credit', 2024, 2),
-- Maxime: paid in full
(4, 3, '2024-01-15', 300.00, 'Cash', 2024, 1),
-- Camille: paid in 4 installments
(5, 4, '2024-02-20', 75.00, 'Debit', 2024, 1),
(6, 4, '2024-05-20', 75.00, 'Debit', 2024, 2),
(7, 4, '2024-08-20', 75.00, 'Debit', 2024, 3),
(8, 4, '2024-11-20', 75.00, 'Debit', 2024, 4),
-- Antoine: paid in full
(9, 5, '2024-06-10', 300.00, 'Cash', 2024, 1),
-- Laurie: paid in 2 installments
(10, 6, '2024-11-01', 150.00, 'Credit', 2024, 1),
(11, 6, '2025-01-01', 150.00, 'Credit', 2024, 2),
-- Gabriel: paid in full
(12, 7, '2024-09-01', 300.00, 'Debit', 2024, 1),
-- Emma: only first installment so far (still owes the rest)
(13, 8, '2024-01-20', 75.00, 'Cash', 2024, 1);

-- =====
-- Table: FIFAGame
-- =====
INSERT INTO FIFAGame (game_id, location_id, team_name, opponent_name, game_date, team_score, opponent_score) VALUES
(1, 1, 'CSCS Head Office', 'Montreal United', '2024-03-10', 3, 1),
(2, 2, 'CSCS West Branch', 'Laval Rovers', '2024-04-05', 2, 2),
(3, 3, 'CSCS East Branch', 'Longueuil FC', '2024-05-12', 1, 4),
(4, 1, 'CSCS Head Office', 'South Shore Kickers', '2024-06-20', 5, 0),
(5, 4, 'CSCS North Branch', 'Pointe-Claire FC', '2024-07-14', 2, 1),
(6, 5, 'CSCS South Branch', 'Brossard United', '2024-08-03', 0, 3),
(7, 2, 'CSCS West Branch', 'Laval Stars', '2024-09-21', 4, 2),
(8, 1, 'CSCS Head Office', 'Outremont Athletic', '2024-10-05', 1, 1),
(9, 3, 'CSCS East Branch', 'Greenfield Park SC', '2024-11-02', 3, 0),
(10, 6, 'CSCS Central Branch', 'Varennnes City FC', '2024-12-07', 2, 2);

-- =====
-- Table: FIFAParticipation
-- =====
INSERT INTO FIFAParticipation (game_id, membership_number) VALUES
(1, 1), (1, 2), (1, 7), -- Game 1: Thomas, Sophie, Gabriel
(2, 3), (2, 4), (2, 8), -- Game 2: Maxime, Camille, Emma
(3, 5), (3, 6), -- Game 3: Antoine, Laurie
(4, 1), (4, 2), (4, 7), -- Game 4: Thomas, Sophie, Gabriel again (repeat participants)
(5, 3), (5, 4), (5, 8), -- Game 5: Maxime, Camille, Emma
(6, 5), (6, 6), (6, 9), -- Game 6: Antoine, Laurie, Alexandre
(7, 1), (7, 3), (7, 4), -- Game 7: Thomas, Maxime, Camille

```

```

(8, 2), (8, 6), (8, 10), -- Game 8: Sophie, Laurie, Isabelle
(9, 7), (9, 8), (9, 9), -- Game 9: Gabriel, Emma, Alexandre
(10, 1), (10, 5), (10, 10); -- Game 10: Thomas, Antoine, Isabelle

-- =====
-- Additional seed data for testing Queries 3 and 4
-- =====

-- =====
-- Query 3 test data:
-- Club Members with at least 4 hobbies
-- =====

-- Camille currently has hobbies 1, 2, and 4.
-- Add hobbies 3 and 5, giving her 5 hobbies.
INSERT INTO ClubMemberHobby (membership_number, hobby_id) VALUES
(4, 3),
(4, 5);

-- Laurie currently has hobbies 1 and 3.
-- Add hobbies 2, 4, and 5, giving her 5 hobbies.
INSERT INTO ClubMemberHobby (membership_number, hobby_id) VALUES
(6, 2),
(6, 4),
(6, 5);

-- =====
-- Query 4 test data:
-- =====
INSERT INTO Payment (
payment_id,
membership_number,
payment_date,
amount,
payment_method,
membership_year,
installment_number
) VALUES

-- Camille: major member, owes $200, fully paid => Active
(14, 4, '2025-02-20', 100.00, 'Debit', 2025, 1),
(15, 4, '2025-06-20', 100.00, 'Debit', 2025, 2),

-- Laurie: major by the end of 2025, owes $200,
-- but paid only $100 => Inactive
(16, 6, '2025-03-01', 100.00, 'Credit', 2025, 1),

-- Alexandre: major member, fully paid $200 => Active
(17, 9, '2025-05-10', 200.00, 'Debit', 2025, 1),

-- Isabelle: major member, paid only $100 => Inactive
(18, 10, '2025-04-15', 100.00, 'Cash', 2025, 1);

```


5. Table Verification — SELECT COUNT(*) FROM R

The output of "SELECT COUNT(*) FROM R;" for each relation R that is formed in the database is listed below. This has been done using sql/04_verify.sql on the local development database (which matches the AITS wqc353_1 database schema).

Relation (R)	COUNT(*)
Location	10
LocationPhone	11
Personnel	10
PersonnelAssignment	10
FamilyMember	10
FamilyMemberAssignment	10
ClubMember	10
ClubMemberFamilyRelation	10
Hobby	10
ClubMemberHobby	18
Payment	18
FIFAGame	10
FIFAParticipation	29

Verification Script (04_verify.sql)

```
-- =====
-- CSCS (Country Soccer Club System) - Verification
-- SELECT COUNT(*) FROM R for every relation R in the database
-- =====

USE wqc353_1;

SELECT 'Location' AS table_name, COUNT(*) AS row_count FROM Location
UNION ALL SELECT 'LocationPhone', COUNT(*) FROM LocationPhone
UNION ALL SELECT 'Personnel', COUNT(*) FROM Personnel
UNION ALL SELECT 'PersonnelAssignment', COUNT(*) FROM PersonnelAssignment
UNION ALL SELECT 'FamilyMember', COUNT(*) FROM FamilyMember
UNION ALL SELECT 'FamilyMemberAssignment', COUNT(*) FROM FamilyMemberAssignment
UNION ALL SELECT 'ClubMember', COUNT(*) FROM ClubMember
UNION ALL SELECT 'ClubMemberFamilyRelation', COUNT(*) FROM ClubMemberFamilyRelation
UNION ALL SELECT 'Hobby', COUNT(*) FROM Hobby
UNION ALL SELECT 'ClubMemberHobby', COUNT(*) FROM ClubMemberHobby
UNION ALL SELECT 'Payment', COUNT(*) FROM Payment
UNION ALL SELECT 'FIFAGame', COUNT(*) FROM FIFAGame

UNION ALL SELECT 'FIFAParticipation', COUNT(*) FROM FIFAParticipation;
```